ME 5309 – Mid-Semester Examination: ANSWER SHEET

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Questions attempted: Q1, Q2, Q3, Q4 (Q2 struck out – not to be considered)

Question 1

Given: sensor 1280 px wide, pixel pitch p = 5 μm (square), lens f = 16 mm, standoff Z = 400 mm; force sensor 10 mV/N, 12-bit ADC over 0–5 V; plant $\dot{x} = -2x + u$ with zero-order hold, T = 0.1 s.

(a) Assumptions: pinhole (thin-lens) camera model, lens focused on the belt plane, belt plane perpendicular to the optical axis, no lens distortion, so every pixel images the same length on the belt.

(i) The focal length in pixel units is the physical focal length divided by the pixel pitch: $f_x = f/p = (16\times10^{-3}\text{ m})/(5\times10^{-6}\text{ m}) = 3200$. The metres cancel so the answer is in pixels. **f_x = 3200 px**.

(ii) Width of the sensor = N·p = 1280 × 5 μm = 6400 μm = 6.4 mm. Using similar triangles between the sensor and the belt plane, $W_s/f = FOV/Z$, hence $FOV = W_s \cdot Z/f = (6.4\text{ mm})(400\text{ mm})/(16\text{ mm}) = 160$ mm. **FOV = 160 mm**.

Ground sampling distance = FOV/N = 160 mm / 1280 px = 0.125 mm/px. Cross-check with the pixel-based formula Z/f_x = 400 mm / 3200 px = 0.125 mm/px; both agree, so **1 pixel = 0.125 mm = 125 μm** on the belt.

(b) Assumptions: ideal 12-bit ADC (4096 codes), the full 0–5 V range is used, no offset or gain error, and sensor noise is below one LSB so quantisation sets the limit.

Voltage step: 1 LSB = $V_{FS}/2^N = 5\text{ V}/4096 = 1.2207\times10^{-3}$ V = 1.2207 mV.

The sensor maps force to voltage by V = S·F with S = 10 mV/N, so $\Delta F_{min} = \Delta V/S$ = 1.2207 mV / (10 mV/N) = 0.12207 N. **$\Delta F_{min} \approx 0.122$ N**. (Using 4095 steps instead of 4096 codes gives 0.1221 N, the same at 3 s.f.)

(c) With u held constant over [kT, (k+1)T] (zero-order hold) the variation-of-constants formula gives $x((k+1)T) = e^{AT}x(kT) + \int_0^T e^{A(T-s)}B\,u(kT)\,ds$. Since u(kT) is constant it leaves the integral, so $F = e^{AT}$ and $G = \int_0^T e^{A\sigma}B\,d\sigma = A^{-1}(e^{AT} - I)B$ for invertible A.

Substituting A = −2, B = 1, T = 0.1 s: $F = e^{-0.2} = 0.81873$. $G = (0.81873 - 1)/(-2) = 0.090635$.

Sanity check: the DC gain of the continuous plant is $-A^{-1}B = 0.5$, and the discrete DC gain $G/(1 - F) = 0.090635/0.18127 = 0.5$ ✓.

$x_{k+1} = 0.8187\,x_k + 0.0906\,u_k$

Question 2 [CROSSED OUT BY STUDENT]

~~*Given: J = 0.02 kg·m², b = 0.05 N·m·s/rad, m = 1 kg, ℓ = 0.25 m, g = 9.81 m/s². So mgℓ = 1 × 9.81 × 0.25 = 2.4525 N·m (used throughout).*~~

~~**(a)** $x_1 = \theta$, $x_2 = \dot{\theta}$. $\dot{x}_1 = x_2$; $\dot{x}_2 = (1/J)[\tau - b\,x_2 + mg\ell \sin x_1]$.~~

~~At equilibrium $x_2 = 0$ and $mg\ell \sin\theta_e = -\tau_0$, so $|\tau_0| \le mg\ell = 2.4525$ N·m.~~

~~**(b)** The equilibrium at $\theta_e = 60°$ needs the torque $\tau_e = mg\ell \sin 60° = 2.4525 \times 0.8660 = 2.124$ N·m. Define deviations $\delta x = x - x_e$ with $x_e = [60°, 0]^T$ and $\delta\tau = \tau - \tau_e$; then $\delta\dot{x} \approx A\,\delta x + B\,\delta\tau$ with the Jacobians evaluated at the equilibrium.~~

~~$\partial f_1/\partial x = [0, 1]$. $\partial f_2/\partial x_1 = -(mg\ell/J)\cos x_1 = -(2.4525/0.02)(0.5) = -122.625 \times 0.5 = -61.31\ (s^{-2})$. $\partial f_2/\partial x_2 = -b/J = -0.05/0.02 = -2.5\ s^{-1}$.~~

~~**A = [0, 1; −61.31, −2.5]**, and since $\partial f_2/\partial\tau = 1/J = 50$, **B = [0; 50]**.~~

~~Check: the characteristic equation $\lambda^2 + 2.5\lambda + 61.31 = 0$ gives $\lambda = -1.25 \pm 7.73j$, a lightly damped stable oscillation, as expected for a pendulum hanging near vertical (60° below the downward~~

~~vertical).~~

~~**(c)** At rest $\dot{\theta} = \ddot{\theta} = 0$: $K_p(\theta_d - \theta_{ss}) = mg\ell \sin\theta_{ss} = 2.4525 \sin\theta_{ss}$. At $\theta_d = 90°$ the right side $\neq 0$ while the left side $= 0$, so the joint cannot rest at θ_d.~~

~~With $\sin\theta_{ss} \approx 1$: $\theta_d - \theta_{ss} = 2.4525/50 = 0.00491$ rad = **0.281°**.~~

~~**(d)** Error: $K_p(\theta_d - \theta_{ss}) = 0$ at rest, so $\theta_{ss} = \theta_d$; **error = 0**.~~

~~Pols: yes they change. The graviy term is removed from the dynamics so the effective stiffness changes from $K_p + mg\ell \cos\theta$ to K_p, which moves the poles (the system becomes faster).~~

~~*Note: K_d does not influence the stedy-state angle because $\dot{\theta} = 0$ there.*~~

~~*(Cancelled - please do not evaluate this question.)*~~

Question 3

Given: $A = [0, 1; 0, -2]$, $B = [0; 1]$, $C = [1, 0]$, $f_s = 1$ kHz so the sample time is $T = 1/f_s = 1$ ms.

(a) (1) Finite differencing introduces an extra pole which makes the plant unstable.

(2) It cannot be implemented digitally because the derivative of a sampled signal is not defined.

(b) $\omega_c = 2\pi \times 20 = 125.66$ rad/s, $a = \omega_c T/2 = 0.0628$, Tustin transform.

$y_k = \mathbf{0.0591\,(u_k + u_{k-1}) + 0.8187\,y_{k-1}}$.

The DC gain is 1 sence low-pass filters always have unity gain at DC.

(c) First check observability: $O = [C; CA] = [1, 0; 0, 1]$, full rank, so the observer poles can be placed arbitrarily.

With $L = [\ell_1, \ell_2]^T$ and $C = [1, 0]$, $LC = [\ell_1, 0; \ell_2, 0]$, so $A - LC = [-\ell_1, 1; -\ell_2, -2]$.

Characteristic polynomial: $\det(sI - (A - LC)) = \det[s + \ell_1, -1; \ell_2, s + 2] = (s + \ell_1)(s + 2) - (-1)(\ell_2) = s^2 + (2 + \ell_1)\,s + (2\ell_1 + \ell_2)$.

Both estimator poles at $s = -10$ means the desiered polynomial is $(s + 10)^2 = s^2 + 20\,s + 100$.
Comparing coefficients: s^1: $2 + \ell_1 = 20 \Rightarrow \ell_1 = 18$. s^0: $2\ell_1 + \ell_2 = 100 \Rightarrow \ell_2 = 100 - 36 = 64$.
$L = [18, 64]^T$. Verification: $A - LC = [-18, 1; -64, -2]$ has trace -20 and determinant $(-18)(-2) - (1)(-64) = 36 + 64 = 100$, so the polynomial is $s^2 + 20s + 100 = (s + 10)^2$ ✓.

(d) Advantage of the observer that no filter of this kind can give: the observer is driven by both the measurment y and the model with the actual input u, so it predicts the motion between samples. The velocity estimate therefore has little or no phase lag. The low-pass filter on differenced data, by contrast, always pays a lag of about $1/(2\pi f_c) = 8$ ms for its noise rejection, a trade-off it cannot escape. In addition the observer gives all states, and can be augmented with a disturbance state (e.g. unknown load torque) which no filter on θ can estimate.

Circumstance where the filter is safer: when the plant model is poorly known or changes during operation, for example an unknown payload, varying friction, contact with the environment, or actuator saturation making u differ from the commanded value. The observer propagates the wrong model so its estimat can be biased (and in the worst case unstable with the feedback loop), whereas the model-free filter just smooths what the encoder actually measures.

Sampling at 1 kHz is fast compared with the 10 rad/s observer bandwidth, so discretisation effects are small.

Question 4

Given: plant $\ddot{\theta} + a\dot{\theta} + c\theta = b\,u$ with $a = 2$, $c = 6$, $b = 1$; controller $u = N_{ff}\,r - k_1\theta - k_2\dot{\theta}$.

(a) Substitute the control law into the plant: $\ddot{\theta} + a\dot{\theta} + c\theta = b\,N_{ff}\,r - b\,k_1\,\theta - b\,k_2\,\dot{\theta}$. Moving the feedback terms to the left: $\ddot{\theta} + (a + b\,k_2)\,\dot{\theta} + (c + b\,k_1)\,\theta = b\,N_{ff}\,r$.

Numerically, $a + b\,k_2 = 2 + 2 = 4$ and $c + b\,k_1 = 6 + 2 = 8$, so the characteristic equation is $s^2 + 4s + 8 = 0$. Roots: $s = [-4 \pm \sqrt{(16 - 32)}]/2 = -2 \pm 2j$. They have negative real parts, so the loop is stable; $\omega_n = \sqrt{8} = 2.83$ rad/s and $\zeta = 4/(2 \times 2.83) = 0.707$.

Transfer function from r to θ: $T(s) = b\,N_{ff} / (s^2 + 4s + 8)$. DC gain ($s = 0$): $T(0) = N_{ff}/8$. For unity DC gain, $\mathbf{N_{ff} = 8}$. (Note that this is larger than 1 becuase the gravity-related stiffness c and the feedback k_1 both resist the displacement.)

(b) With the actual $b = 0.75$ and unchanged $N_{ff} = 8$, $k_1 = k_2 = 2$, the closed loop becomes $\ddot{\theta} + (a + bk_2)\dot{\theta} + (c + bk_1)\theta = bN_{ff}\,r$, i.e. $\ddot{\theta} + 3.5\dot{\theta} + 7.5\theta = 6\,r$.

For a unit step the steady state is found by setting derivatives to zero: $7.5\,\theta_{ss} = 6 \times 1$, so $\theta_{ss} = 6/7.5 = \mathbf{0.8}$.

Tracking error $= (r - \theta_{ss})/r \times 100\% = (1 - 0.8) \times 100 = \mathbf{20\%}$. A common mistake would be to say the output falls by the same 25% as the acuator gain; that is wrong here, because b also appears in the feedback stiffness $c + bk_1$, which decreases from 8 to 7.5 and partially compensates. The pols also change: $s^2 + 3.5s + 7.5 \rightarrow s = -1.75 \pm 2.11j$, still stable.

(c) Take the state $z = [\theta, \dot{\theta}, \xi]^T$. The plant gives $\ddot{\theta} = -a\dot{\theta} - c\theta + b\,u$ with $u = -k_1\theta - k_2\dot{\theta} + K_I\xi$, and $\dot{\xi} = r - \theta$. With $r = 0$ (r only shifts the equilibrium) the closed-loop matrix is $A_{cl} = [0, 1, 0; -(c + bk_1), -(a + bk_2), bK_I; -1, 0, 0]$.

Characterstic polynomal $\det(sI - A_{cl})$, expanding along the first row: $s\cdot[(s + a + bk_2)s - 0] + 1\cdot[(c + bk_1)s + bK_I] = s^3 + (a + bk_2)s^2 + (c + bk_1)s + bK_I$. **(Result)**

Desiered poles -4 and $-2 \pm 2j$: $(s + 2 - 2j)(s + 2 + 2j) = (s + 2)^2 + 4 = s^2 + 4s + 8$. Multiply by $(s + 4)$: $s^3 + 4s^2 + 8s + 4s^2 + 16s + 32 = s^3 + 8s^2 + 24s + 32$.

Compare coefficients with $a = 2$, $b = 1$, $c = 6$: s^2: $2 + k_2 = 8 \Rightarrow \mathbf{k_2 = 6}$. s^1: $6 + k_1 = 24 \Rightarrow \mathbf{k_1 = 18}$. s^0: $b\,K_I = 32 \Rightarrow \mathbf{K_I = 32}$.

Chek: all coefficients of the desiered polynomial are positive and $(8)(24) = 192 > 32$, so Routh–Hurwitz confirms the three poles are in the left half plane ✓.

(d) Suppose the closed loop is stable and r is constant. Then the state converges to a constant, so $\dot{\theta} = \ddot{\theta} = 0$ and, crucially, $\dot{\xi} = 0$ as well (ξ itself is constant at its final value ξ_{ss}). From the integrator equation $\dot{\xi} = r - \theta$, we get $r - \theta_{ss} = 0$, i.e. $\boldsymbol{\theta_{ss} = r}$.

Neither b nor c appears in this argument. They only influence the value of ξ_{ss} needed to hold the equilibrium (from the plant equation $c\theta_{ss} = bu_{ss} = b(-k_1\theta_{ss} + K_I\xi_{ss})$), and the pole locations, so the zero error survives any error in the model parameters provided the loop stays stable (unlike the feed-forward N_{ff} scheme of part (b)).

Practical failure mode: integrater windup. If the acuator saturates (torque limit), the loop is effectively open, the error stays non-zero and ξ keeps integrating to a huge value. When the reference becomes reachable the stored integral forces a large overshoot and a long recovery. Standard remedy: anti-windup, such as clamping the integrater when the actuator is saturated, conditional integration, or back-calculation that feeds the difference between commanded and saturated torqe back to the integrator.